1 — Installing SDOM (Step-by-Step Tutorial)

This tutorial walks you through installing the **Storage Deployment Optimization Model (SDOM)** on a Windows machine. It is based on the official SDOM documentation:

- SDOM Documentation home: https://natlabrockies.github.io/SDOM/index.html
- Developer / environment setup guide: https://natlabrockies.github.io/SDOM/sdom_Developers_guide.html
- SDOM GitHub repository: https://github.com/Omar0902/SDOM

> SDOM is an open-source, Python/Pyomo-based, high-resolution capacity-expansion framework developed by the National Lab of the Rockies (NLR). See the SDOM home page for an overview.

Table of Contents

---

1. Prerequisites

Before starting, make sure you have:

- A Windows, macOS or Linux machine with administrator rights.
- An internet connection (to download Python, `uv`, and SDOM).
- ~2 GB of free disk space (Python + virtual env + solver).

Reference: SDOM Installation — System Setup and Prerequisites

---

2. Install Python

1. Download Python (3.10 or newer is recommended) from the official site: https://www.python.org/downloads/

2. Run the installer. **Important**: on the first installer screen, check the box **"Add Python to PATH"**.

3. Confirm Python is on your `PATH`. If not, follow this guide to add it: https://realpython.com/add-python-to-path/

4. Open a new PowerShell window and verify:

```
python --version
pip --version
```

You should see Python and pip versions printed.

---

3. Install VS Code and recommended extensions

SDOM docs recommend Visual Studio Code as the IDE.

1. Download and install VS Code: https://code.visualstudio.com/
2. Install these recommended extensions from the VS Code Marketplace:
   - **Python** (Microsoft)
   - **edit CSV** — to edit SDOM input CSVs directly in VS Code: https://marketplace.visualstudio.com/items?itemName=janisdd.vscode-edit-csv
   - **vscode-pdf** — to view PDFs inside VS Code: https://marketplace.visualstudio.com/items?itemName=tomoki1207.pdf

---

4. Install `uv` (Python environment manager)

SDOM uses `uv` to manage virtual environments and dependencies.

Open PowerShell and run:

```
pip install uv
```

Verify:

```
uv --version
```

Reference: Install uv

---

5. Create a virtual environment

Choose (or create) a working folder for your SDOM project, `cd` into it, and create a virtual environment named `.venv`:

```
# Move to your project folder
cd C:\Users\<your-user>\repos\<your-project>

# Create the virtual environment
uv venv .venv

# Activate it (Windows PowerShell)
.venv\Scripts\Activate.ps1
```

For macOS / Linux:

```
source .venv/bin/activate
```

When activation succeeds the prompt is prefixed with `(.venv)`.

> If PowerShell blocks script activation, run once (as Administrator): `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`

---

6. Install SDOM

You have two ways to install SDOM. Pick the one that matches your use case.

With the `.venv` activated:

```
uv pip install sdom
```

---

7. Install a solver (CBC / HiGHS)

SDOM is a Pyomo model and needs an external (or bundled) MILP solver.

- **HiGHS** is installed automatically as a dependency through the `highspy` Python package — no extra step required.
- **CBC** (used in many SDOM examples) is provided as a binary inside the `Solver/bin/` directory of the SDOM repo (e.g. `./Solver/bin/cbc.exe`). When installing from source (Option B), that folder is already available.
- If you used Option A (PyPI), and you want to use CBC, download a CBC binary from https://github.com/coin-or/Cbc/releases and remember its path — you will reference it in `get_default_solver_config_dict(executable_path=...)`.

---

8. Verify the installation

With the virtual environment activated:

```
python -c "import sdom; print(sdom.__version__)"
```

You should see a version number printed (no `ImportError`).

---

9. Quick Start example

A minimal SDOM run looks like this (from the official Quick Start):

```python
from sdom import (
    load_data,
    initialize_model,
    run_solver,
    get_default_solver_config_dict,
    export_results,
)

# Load input data
data = load_data("./Data/your_scenario/")

# Initialize model (8760 hours = full year)
model = initialize_model(data, n_hours=8760)

# Configure solver (CBC example)
solver_config = get_default_solver_config_dict(
    solver_name="cbc",
    executable_path="./Solver/bin/cbc.exe",
)

# Solve
results = run_solver(model, solver_config)

if results.is_optimal:
    print(f"Total Cost: ${results.total_cost:,.2f}")
    print(f"Wind Capacity: {results.total_cap_wind:.2f} MW")
    print(f"Solar Capacity: {results.total_cap_pv:.2f} MW")
    export_results(results, case="scenario_1", output_dir="./results/")
```

For details on inputs and outputs, see:

- SDOM Input Data
- Running SDOM and Understanding Outputs

---

10. Troubleshooting

- **`uv` is not recognized** — Re-open PowerShell after `pip install uv`, or check your `PATH`.
- **Activation script is blocked** — run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` in PowerShell.

- `ImportError: sdom` — make sure the `.venv` is **activated** before running Python; the prompt should start with `(.venv)`.
- **Hardlink warning during `uv pip install`** — harmless; you can ignore it or set `set UV_LINK_MODE=copy`.
- **Solver not found** — pass an explicit `executable_path` to `get_default_solver_config_dict(...)` pointing to your CBC/HiGHS binary.
- See also: Troubleshooting.

---

11. References

- SDOM Documentation (home)
- Installation section
- Developer / environment setup guide
- SDOM GitHub repository
- User Guide — Introduction
- User Guide — Input Data
- User Guide — Running SDOM and Outputs
- API Reference
- `uv` on PyPI
- Python downloads
- VS Code